NudgeWell

Personalized prediction of unwanted continuing smartphone sessions

Digital Addiction Idea Proposal | Group 6 | 24 September 2026

1 Project Idea

Many phone warnings fire after a fixed time, even during purposeful use. NudgeWell predicts the session-level intention-behavior gap: whether a user who feels they should stop is likely to keep using the phone. It will use only signals available during the session, such as time, app category, elapsed use, interactions, and earlier phone use. A high-confidence alert in our Android prototype would offer a short activity chosen from the user's stated preferences, with an option to dismiss it. Deliverables are the prediction model, evaluation, and app demo. The estimate is not an addiction diagnosis.

2 Literature Examples

Study and evidence	Contribution to NudgeWell
Zheng and Choe (2026) *To Stop or Not to Stop* 5,772 sessions | 37 users | doi:10.1145/3821659	Supplies the exact session-level target and a random-forest benchmark; its 37-person cohort limits generalization.
Orzikulova et al. (2024) *Time2Stop* 8 weeks | 71 users | doi:10.1145/3613904.3642747	Motivates feedback-based timing and user control; it does not validate NudgeWell's predictions or interventions.

3 Describe Your Data

Primary training data: AwareMind (osf.io/x6yzs), with 5,772 labelled in-session reports from 37 adult Android participants. Each report records a 1-10 stop/continue intention; subsequent remaining use yields the session-level gap. The study recorded 31,349 phone sessions overall, but only 5,772 have these labels. We will audit released fields, use only prediction-time inputs, and confirm data-reuse permission before training. Secondary data: one sec (osf.io/bmjat) records 1,318,331 app-opening intervention events from 1,039 users. Its events have no intention-gap labels and may support only a separate response analysis. CMI measures person-level risk and will not train this session-level predictor.

4 Approach

We will implement a random-forest regression comparable to the study's benchmark and simple baselines. If timestamps and usable histories permit, we will test a compact temporal model using only earlier sessions and current pre-prompt signals. The primary output is the continuous intention-behavior gap; separate intention and remaining-use models are optional diagnostic comparisons. We will hold out participants for new-user evaluation and test personalization chronologically. Metrics include gap MAE and R-squared, plus high-gap precision-recall and simulated alert burden at a predeclared threshold. With 37 participants, overfitting is a serious risk; we will report uncertainty and compare honestly with simpler baselines. Suggestions follow stated preferences; effectiveness requires a later user study.